Diagnosis — G1 EDU "active mode" failure

Date: 2026-10-26

Context: after a fall (2026-08-26) and replacement of the right thigh/knee actuator, the robot enters ready mode normally but diverges violently as soon as it switches to active mode, and drops back into damp.

Conclusion (proven by the data)

> At rest, the state estimator is healthy (velocity = 0.000 m/s, verified with the robot powered on and standing still). On every switch to active mode, it re-initialises with a velocity of about 0.15 m/s with no physical cause whatsoever: at the exact instant of the jump the IMU is quiet, the legs are motionless, and the controller has not yet taken over. The balance controller believes this false velocity, commands a step to "catch up", and the feedback loop runs away to divergence (up to 1.3 m/s linear, 10 rad/s in yaw).

It is neither the IMU (replaced -> no change), nor a mechanical knee lock (legs motionless at the instant of the fault; right-knee travel, friction and end-stops measured identical to the left), nor a static joint offset (invisible at rest, and the heading of the error is not constant — see below).

Evidence: the switch instant, cycle by cycle

File g1_guardian_data_20261026_114812_event.npz (6000 samples, ~500 Hz). Estimated velocity is zero for 8 s, then jumps at t ≈ 8.15 s. Window around the jump (t0 = 8.11 s):

dt (s)	|v|est (m/s)	gyro (rad/s)	dq_L max	dq_R max	leg kp	state
0.032	0.000	0.01	0.02	0.03	1600	motionless, ready
0.035	0.153	0.01	0.02	0.03	1600	<- phantom velocity: nothing moves, controller not engaged
0.085	0.160	0.01	0.02	0.02	1080	controller engages (kp 1600->1080)
0.089	0.160	0.01	0.53	0.18	1080	legs start to move
0.096+	0.196 -> 3.0	3 -> 12	thrash	thrash	1080	run-away -> guardian cut

Decisive points:

- The phantom velocity precedes any physical motion and precedes the controller's engagement by ~70 ms. It is born at the instant the estimator switches its computation mode, not from a sensor.
- The estimator's errcode stays 0 throughout: it is not aware of being wrong, so no internal safeguard stops it.

The fault is not a fixed bias (so not fixable by an external offset)

Heading (vx, vy) of the phantom velocity at its birth, across 11 runs with a real engagement (angle: 0° = forward, +90° = left, −90° = right):

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-42   -15   +102   -158   -168   -29   +18   +10   +31   +1   -33     (degrees)
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- Magnitude always ~0.1–0.3 m/s (never zero) -> a systematic perturbation in size.
- Random direction from one run to the next -> NOT a fixed encoder offset (otherwise the heading would be constant).
- First leg to move: right 5x, left 2x, ~equal 4x -> "always the right knee" is false.

Signature of an estimator re-initialisation from a state that depends on the micro-posture at the switch instant, not of a localised mechanical error.

Likely origin (link with the incident)

Replacing the actuator introduces a different encoder zero / leg parameter that the robot has never rewritten (a commissioning step was skipped). At rest the estimator has nothing to recompute -> it looks healthy. On the switch, it recomputes its reference from a right-leg geometry inconsistent with the left -> position jump -> derivative = phantom velocity. The variability of the heading comes from the reconciliation depending on the exact posture at the moment of the switch.

This is why neither joint calibration (Explore app) nor IMU replacement changed anything: neither of them rewrites the link between the physical actuator fitted and the estimator's odometry model.

Solution

Only true fix: redo the demarcate (biao-ding) / re-registration of the right actuator in the robot, so the estimator initialises on a consistent geometry. This is the commissioning step missing after the part swap.

Limitation: the basic_demarcate service runs on the closed control board (192.168.123.161) — no SSH, no TCP port, no client in the public SDK. Triggering it goes through Unitree's internal tool/procedure.

External workaround (fixed joint offset via the app): ruled out. It would only work if the error were a constant bias; but the heading changes every run, so a fixed offset would fix one run and not the next.

What was ruled out experimentally

Hypothesis	Verdict	Evidence
IMU mounted wrong / faulty	ruled out	IMU replaced -> phantom velocity unchanged; IMU quiet at the fault instant
Shifted joint zero (static)	ruled out as direct cause	invisible at rest; heading of the error not constant
Right knee locked / mechanical	ruled out	legs motionless at the fault instant; right travel/friction/end-stops = left
Joint calibration	no effect	done -> no change
Arms along the body / posture	ruled out	ready posture correct and symmetric

Figures

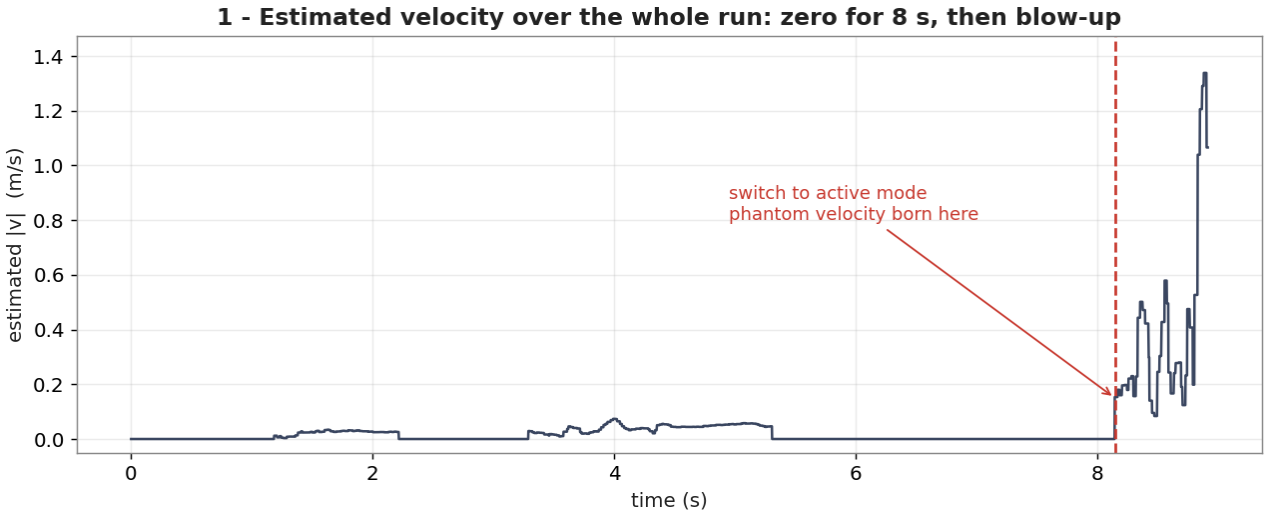


Fig. 1 — Estimated velocity over the whole run: flat at zero for 8 s, then blow-up at the switch.

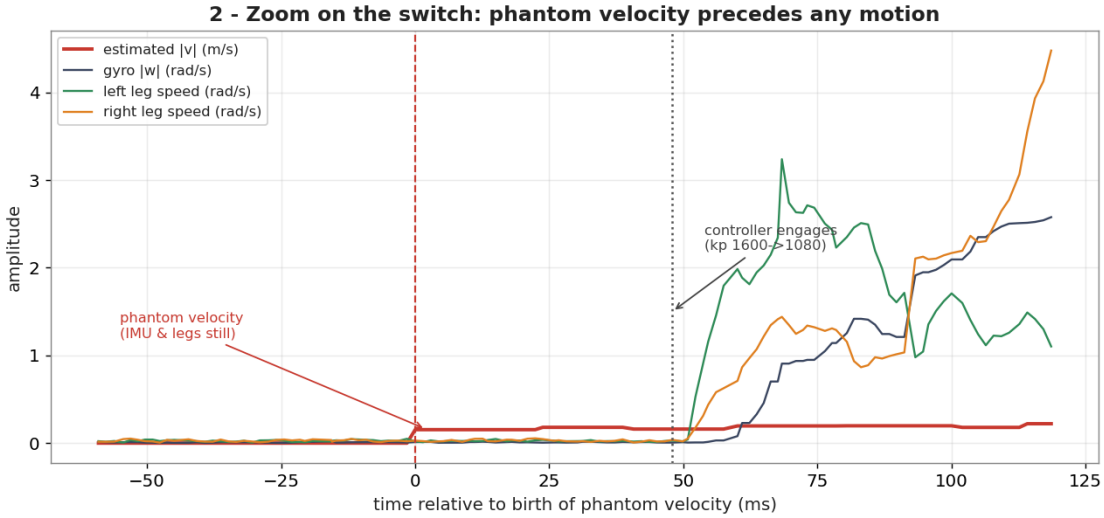


Fig. 2 — Zoom on the switch: the phantom velocity (red) jumps while gyro and both legs stay at zero; physical motion starts ~50 ms later, when the controller engages.

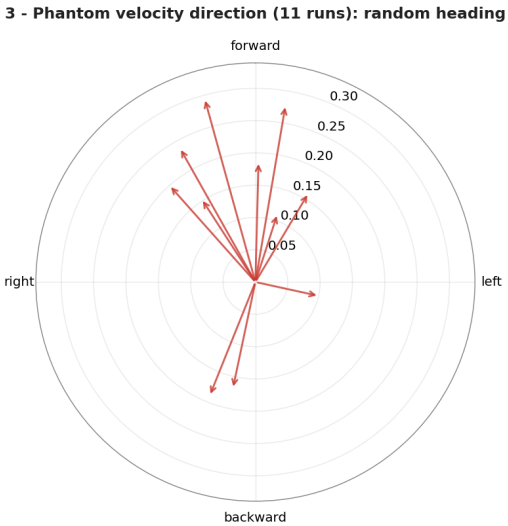


Fig. 3 — Phantom velocity heading over 11 runs: arrows point in all directions -> not a fixed bias.

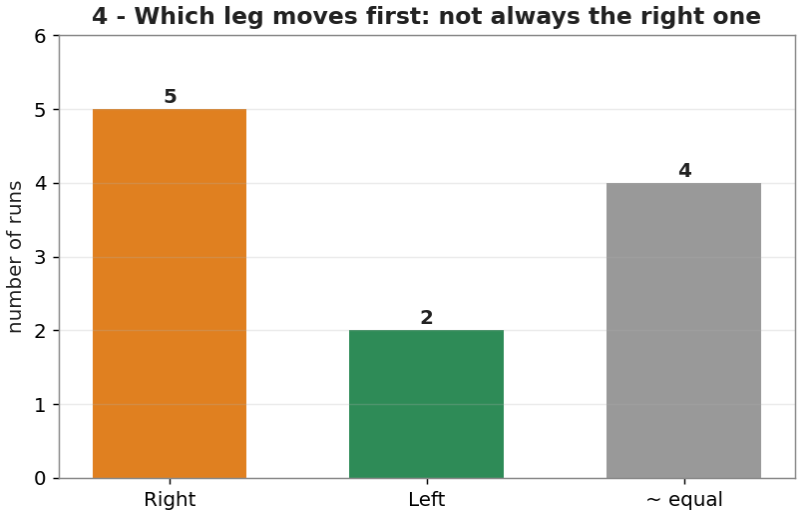


Fig. 4 — First leg to move: right 5x, left 2x, equal 4x -> not always the right one.